

The Logistic Regression Model

- **Logistic Regression** is one of the most widely used classification algorithms.
 - Unlike linear regression, which fits a straight line, logistic regression fits an **S-shaped curve** to the data. This curve is more suitable for classification tasks where the output is either 0 or 1.
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The Sigmoid (Logistic) Function

- The core of the logistic regression model is the **Sigmoid function**, denoted as $g(z)$.

- **Formula:**

$$g(z) = \frac{1}{1+e^{-z}}$$

where e is the mathematical constant ≈ 2.718 .

- **Properties of the Sigmoid Function:**

- It takes any real number z as input and outputs a value between **0 and 1**.
 - If z is a large positive number, e^{-z} approaches 0, so $g(z)$ approaches 1.
 - If z is a large negative number, e^{-z} becomes very large, so $g(z)$ approaches 0.
 - When $z = 0$, $g(0) = \frac{1}{1+e^0} = \frac{1}{1+1} = 0.5$.
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Constructing the Model

The logistic regression model is built in two steps:

1. **Linear Combination:** First, calculate z using the same linear function as in regression:
$$z = \vec{w} \cdot \vec{x} + b$$
2. **Apply Sigmoid Function:** Pass the result z through the sigmoid function to get the final prediction:

$$f_{\vec{w},b}(\vec{x}) = g(z) = g(\vec{w} \cdot \vec{x} + b)$$

- **Full Model Equation:**

$$f_{\vec{w},b}(\vec{x}) = \frac{1}{1+e^{-(\vec{w} \cdot \vec{x} + b)}}$$

Interpreting the Output 🤖

- The output of the logistic regression model, $f_{\vec{w},b}(\vec{x})$, is interpreted as the **probability** that the output label y is equal to 1, given the input features $\vec{x}$.
- **Formal Notation:**
$$f_{\vec{w},b}(\vec{x}) = P(y = 1 | \vec{x}; \vec{w}, b)$$
- **Example:**

- If the model outputs 0.7 for a given tumor size, it means there is a **70% probability** that the tumor is malignant ($y = 1$).
- Since the probabilities must sum to 1, the probability of the tumor being benign ($y = 0$) is:
$$P(y = 0) = 1 - P(y = 1) = 1 - 0.7 = 0.3 \quad (30\% \text{ probability})$$
- The next step in using the model is to define a **decision boundary**, which determines how to map these probability outputs (e.g., 0.7) to a final, discrete class prediction (0 or 1).